Interpretable Action-Item Detection in Long Meeting Transcripts Using Linear SVM and Linguistic Feature Modeling

Venus Ruselle B. Daanoy

College of Informatics and Computing Studies, New Era University, venusruselle.daanoy@neu.edu.ph

John Keith S. Mercado

College of Informatics and Computing Studies, New Era University, johnkeith.mercado@neu.edu.ph

ABSTRACT

The rapid shift toward hybrid work environments has intensified the challenge of "information overload," particularly in generating accurate Minutes of the Meeting (MoM) for multilingual dialogues. In the Philippine context, the prevalence of code-switching (Taglish) in professional and academic settings presents a significant barrier for standard Natural Language Processing (NLP) models, which often lack the transparency required for critical task tracking. This study presents a high-precision action item extraction system utilizing a Linear Support Vector Machine (SVM) framework. To address the linguistic complexity of Filipino-English discourse, we developed a massive, diverse corpus consisting of 180,000 samples spanning business, healthcare, education, and legal domains.

Our methodology employs Feature Hashing for 2^{10} vector mapping and a Stochastic Gradient Descent (SGD) training approach to ensure scalability and model interpretability. Experimental results demonstrate that the linear decision function effectively identifies linguistic markers unique to Taglish imperatives (e.g., *"paki-send"* or *"need to finish"*), achieving robust F1-scores across all four domains. A comparative analysis reveals that increasing the data scale from 10k to 180k significantly stabilizes precision, providing a transparent alternative to "black-box" deep learning models. This research concludes that a well-optimized Linear SVM, paired with a large-scale multilingual dataset, offers an efficient and interpretable solution for automated meeting documentation in code-switching environments.

CCS CONCEPTS • Computing Methodologies – Supervised learning by classification: Support vector machine • Computing Methodologies – Natural language processing: Information extraction; • Information Systems- Document summarization.

Additional Keywords and Phrases: *Action Item Extraction, Taglish NLP, Linear SVM, Feature Engineering, Minutes of the Meeting, Multilingual Text Classification..*

1 INTRODUCTION

The rapid transition to hybrid and remote work environments has led to a significant increase in the volume of digital meetings conducted across various sectors, including business, academia, and healthcare. While video conferencing tools provide accessibility, they also generate a massive amount of unstructured conversational data. Distilling these long-form

transcripts into concise, actionable "Minutes of the Meeting" (MoM) remains a labor-intensive task that is prone to human error and information loss.

A critical component of effective meeting summarization is Action-Item Detection—the process of identifying specific tasks, assignments, or commitments made during a discussion. However, automating this process presents two primary challenges. First, existing Natural Language Processing (NLP) models often struggle with "Taglish," a common linguistic phenomenon in the Philippines where English and Tagalog are used interchangeably. Second, many modern "black-box" architectures, such as deep neural networks, lack the interpretability required for users to understand the logic behind task extraction.

This study addresses these challenges by proposing an interpretable action-item detection system using a Linear Support Vector Machine (SVM). By leveraging Linguistic Feature Modeling (LFM), the system identifies imperative constructs and domain-specific markers without the heavy computational requirements of transformer-based models. To ensure the model's robustness across diverse professional fields, we utilize an expanded corpus of over 180,000 samples, covering specialized sectors such as Legal, Engineering, and Healthcare. This research aims to provide a high-precision, transparent, and scalable framework for automated task management in linguistically diverse environments.

1.1 Background of Study

In professional and organizational settings, meetings serve as a central mechanism for communication, collaboration, and decision-making. They are critical for assigning tasks, discussing project directions, and resolving issues. However, long and unstructured meeting transcripts—particularly in hybrid and remote work environments—pose challenges in tracking important information, especially action items. These action items, which often represent assigned tasks or decisions requiring follow-up, are frequently buried within lengthy dialogues, making them difficult to identify and retrieve efficiently without automated support.

To address this, the field of natural language processing (NLP) has explored various techniques for meeting summarization and action-item detection. Many early systems have focused on abstractive or extractive summarization, aiming to condense key points of a meeting transcript. However, summarization alone does not suffice when the goal is to isolate action-oriented utterances. Thus, action-item detection has emerged as a specialized task within the broader field of dialogue understanding.

Recent advancements in NLP, particularly transformer-based models such as BERT and its derivatives, have significantly improved performance in text classification and summarization tasks. For example, Golia and Kalita (2024) employed a hybrid extractive-abstractive summarization pipeline using BERTScore to identify action points in meetings with high accuracy. However, these models are computationally intensive, less interpretable, and often require fine-tuning with large, domain-specific datasets—limitations that make them less practical for low-resource environments or for deployment in real-time systems.

Further, the work by Kiminos and Daskalaki (2020) compared Naïve Bayes and SVMs for classifying useful information in business meetings and found that SVMs offered better precision for decision and task-related utterances, particularly when feature engineering emphasized domain-relevant keywords, modal verbs, and syntactic patterns (Kiminos & Daskalaki, 2020). This affirms the potential of combining domain-specific linguistic features with classical models to develop efficient, interpretable, and scalable systems for action-item detection. Moreover, the IEEE study by Xie et al. (2023) on regularized context modeling provided further support for leveraging structured context and lexical semantics in classical ML pipelines, improving generalizability across meetings with varied formats (Xie et al., 2023).

Building upon this, the current study seeks to apply Linear SVM modeling integrated with engineered linguistic features to detect action items in long meeting transcripts. Rather than relying solely on deep contextual embeddings, this approach emphasizes syntactic and semantic cues found in real human dialogue, such as imperative phrasing, modals (e.g., “should”, “need to”), named entity references, and discourse markers. This not only enhances transparency and interpretability but also allows for greater control over model behavior in culturally nuanced scenarios.

The study further emphasizes local adaptation by integrating the AMI Meeting Corpus, a well-established multilingual meeting dataset, and the i-TANONG dataset, which includes authentic Philippine-based meeting dialogues in English, Tagalog, and Cebuano. This combination enables the system to generalize well across varied linguistic environments while ensuring relevance to hybrid work settings in the Philippines—where code-switching, politeness strategies, and indirect speech are more prevalent.

By focusing on action-item detection rather than full summarization, this research directly targets usability and workflow integration in task management systems, project collaboration tools, and productivity software used by Filipino professionals. It also lays the groundwork for extending such systems to other regional languages and settings, promoting inclusivity in NLP applications.

1.2 Statement of the Problem

In the shift toward remote and hybrid work, organizations have become heavily reliant on digital meetings. While this ensures continuity, it has resulted in an overwhelming number of long-form transcripts that are difficult to manage. Most actionable tasks and decisions are buried within hours of dialogue, leading to “information overload” and missed deadlines. Existing Natural Language Processing (NLP) solutions often struggle with the informal nature of spoken conversation and the high computational cost of transformer models. Furthermore, most systems are optimized for Western discourse, failing to capture the unique code-switching and politeness strategies used by Filipino professionals. Specifically, this study will seek to answer the following questions:

1. How will the integration of linguistic feature modeling (e.g., modal verbs, dependency relations, and discourse markers) improve the detection accuracy of action items in long meeting transcripts compared to baseline models?
2. What is the benefit of using a Linear SVM model over deep learning "black-box" architectures in terms of model interpretability for researchers and meeting participants?
3. How effective is the system in a localized context, specifically in identifying action items that involve Taglish (Tagalog-English) code-switching and Filipino-specific politeness markers (e.g., *"paki," "po"*)?
4. How efficient is the proposed Action-Item Detection System in terms of:

   a. Precision and Recall: Its ability to correctly identify tasks without flagging general conversation.
   b. Interpretability: The clarity of "why" a sentence was classified as an action item.
   c. Generalizability: Its performance across different datasets (AMI vs. i-TANONG).
   d. Contextual Clarity: The readability of the final summarized action items.

1.3 Objectives

The Interpretable Action-Item Detection system aims to attain the following objectives:

1.3.1 Main Objectives

The main objective is to develop an action-item detection system for long meeting transcripts using a linear Support Vector Machine (SVM) model, while adding linguistic and contextual features, aiming to accurately identify actionable statements and enhance usability in terms of contexts of hybrid work in the Philippines.

1.3.2 Specific Objectives:

1. To preprocess long-form meeting transcripts from both the AMI Corpus and i-TANONG datasets by applying cleaning techniques such as text normalization, speaker turn segmentation, and sentence-level structuring, in order to prepare consistent and high-quality inputs for action-item detection.
2. To extract and model linguistically relevant features, including dependency relations, discourse markers, sentence position, and verb modality, which could provide insight into action item identification from most party conversations.
3. To develop and train a Linear Support Vector Machine (SVM) classifier using the linguistically relevant features created and optimize the parameters of the model utilizing cross-validation to attain maximum precision and recall without deteriorating model interpretability.
4. To evaluate the trained model's effectiveness in detecting action items using standard performance metrics across both the AMI and i-TANONG datasets, while focusing on evaluating the models performance in a culturally localized Filipino meeting setting.
5. To integrate the trained linear SVM model into a fully functional action-item detection system that could process long meeting transcripts and produce role-agnostic predictions, and display extracted action items in an organized format that is ready for practical use in hybrid work contexts suitable for practical deployment in hybrid work settings.

2 SCOPE AND LIMITATIONS

The primary focus of this research is the development of an interpretable action-item detection system and a minutes-of-the-meeting generator tailored for long, unstructured meeting transcripts using a Linear Support Vector Machine (SVM). The system utilizes Linguistic Feature Modeling, targeting imperative verbs, modals, and discourse markers, to identify actionable tasks within long transcripts. Training and evaluation are conducted using the *AMI Meeting Corpus* and the *i-TANONG* dataset, specifically covering English and Tagalog within the Philippine hybrid work context. The final output is a structured Minutes of the Meeting (MoM) report containing extracted tasks, assignees, and deadlines.

Conversely, certain boundaries and constraints are recognized in this study. The research is delimited to text-based analysis; thus, it does not include the development of a Speech-to-Text (STT) engine and relies on the assumption that input transcripts are of sufficient quality. Furthermore, the system focuses on extractive summarization techniques for the generation of meeting minutes, meaning it will not perform abstractive summarization or paraphrase dialogues into new text. In terms of linguistic coverage, while the study includes major Philippine languages, it does not account for other regional dialects or highly specialized technical jargon outside the scope of the provided datasets. Methodologically, as the study utilizes a Linear SVM rather than deep learning architectures like Transformers, the system may face limitations in capturing long-range contextual dependencies that span across distant parts of a long transcript. Finally, the system is designed for asynchronous processing of completed transcripts rather than real-time, live-streamed action-item detection

3 RELATED WORK

The automated analysis of meeting transcripts has seen significant growth with the rise of remote collaboration. This section reviews the evolution of action-item detection and summarization, specifically focusing on the transition from traditional linguistic models to machine learning approaches, and the unique challenges posed by multilingual environments.

3.1 Automatic Text Summarization

Automatic text summarization (ATS) aims to distill documents into shorter versions while keeping the key information and maintaining the same meaning. Critical reviews of the current state of ATS research identify two main approaches to summarization, extractive (taking the salient sentences) and abstractive (generate summaries that paraphrase). Widyassari et al. (2022) provide a review of over 85 methods for ATS from 2008-2019. The authors conclude there's solid performance on very short texts; however, many of those methods were unable to maintain contextual coherence or factual consistencies with longer, conversational data.

A broader view of summarization methodologies is offered by Yadav et al. (2022), who conducted a systematic meta-analysis on both extractive and abstractive summarization techniques. Their study categorizes various models based on preprocessing strategies, text representations, and evaluation metrics. Extractive methods, which select existing sentences from a source text, are noted for their efficiency, while abstractive techniques generate novel phrases and require deeper semantic understanding.

Patel and Pareek (2021) provide a comprehensive overview of the evolution, advancements, and remaining challenges in automatic text summarization, with particular emphasis on deep learning–based methods. Their study charts the transition from statistical and rule-based systems to neural networks, such as sequence-to-sequence models and transformer architectures. A key issue identified is factual consistency, wherein abstractive models may hallucinate or generate incorrect content. Scalability and generalization across domains also remain pressing concerns.

Finally, Zhang et al. (2020) reconceptualize extractive summarization as a text matching task, where candidate sentences are aligned against a semantic representation of the source document. By reframing the summarization process in this retrieval-like structure, their approach leverages *neural matching architectures* that improve the selection of relevant and salient sentences. This shift toward sentence-level matching helps mitigate common limitations of traditional extractive systems, such as inclusion of non-essential or off-topic content. In the context of meetings, where actionable insights are often dispersed across varied dialogue turns, such matching-based strategies could enhance the focus and precision of automated summaries.

3.2 Action-Item Detection with Feature Engineering & SVM

Action-item detection has become an essential task in automatic meeting analysis, as it allows systems to identify and extract actionable outcomes from conversational transcripts. Stojanovic et al. (2023) introduced an approach that leverages regularized context modeling to detect meeting action items from transcripts, emphasizing the importance of contextual features such as nearby dialogue acts and sentence types. Their model integrates handcrafted features with Support Vector Machines (SVMs), showing that the inclusion of regularized context windows significantly boosts the performance of traditional machine learning models.

Gishamer et al. (2023) explored action item detection within multiparty conversational speech by analyzing meeting transcripts through a system that integrates domain-specific features. The researchers emphasized the challenges of

overlapping speech, ambiguous action phrasing, and varied speaker intents, and addressed them through an enriched feature set and fine-tuned SVM decision boundaries.

Kiminos et al. (2020) conducted a comparative study between Naïve Bayes and Support Vector Machines for classifying decisions in business meeting transcripts. Although both models aimed at identifying meaningful segments such as decisions and tasks, SVM demonstrated superior performance due to its ability to model high-dimensional feature spaces and handle sparse data. Their feature engineering pipeline included term frequency-inverse document frequency (TF-IDF), cue phrases, and syntactic structures, which significantly improved the precision of action item detection.

In broader research related to conversational structure recognition, Li et al. (2021) examined the role of punctuation, labels, and context in helping transformers and other models identify dialogue acts. While their focus was not directly on SVMs, their findings supported the underlying premise that contextual and structural features are critical for understanding speaker intent. Their conclusions offer indirect validation for the feature engineering strategies employed in SVM-based models.

3.3 Linguistic and Contextual Features

Understanding the role of linguistic features in Natural Language Processing (NLP) tasks has proven critical, especially in applications like meeting summarization and action-item detection. One pivotal study by Tenney et al. (2019) explored how contextual word representations—such as those derived from models like BERT—can effectively capture various layers of linguistic knowledge including syntax, semantics, and coreference resolution. Their findings demonstrated that deep language models inherently learn transferable linguistic patterns that benefit downstream tasks such as sentence classification, entailment detection, and question answering. In the context of meeting transcripts, where utterances are often incomplete or informal, leveraging such contextual representations can help identify nuanced linguistic cues that distinguish actionable statements from general dialogue.

Further expanding on linguistic intricacies, the work of Crossley et al. (2020) provided a comprehensive overview of how linguistic features contribute to writing quality and textual coherence. While their research focused on writing development, their framework of analyzing lexical diversity, syntactic complexity, and cohesion can also be adapted to spoken dialogue in meetings. Transcripts often exhibit varying levels of coherence and structure depending on the speaker and context. Recognizing and modeling such features can significantly enhance the identification of task-oriented utterances or action items, especially when differentiating between vague suggestions and clearly defined directives.

On the other hand, contextual modeling combined with syntactic analysis has also been applied in sentiment analysis, which shares structural similarities with action-item detection. In their study, Wu et al. (2020) proposed a hybrid model that integrated contextual embeddings with syntactic features to improve aspect-based sentiment analysis. Their approach emphasized the importance of understanding the relationship between sentence elements to derive specific insights. In a similar vein, action-item detection can benefit from this method by focusing not just on isolated keywords but also on how clauses are structured—identifying intents, subjects, and corresponding actions from nested or compound sentences often found in meetings.

Lastly, the recent work by Yadav et al. (2024) introduced a dialogue-act–enhanced framework for meeting summarization, showcasing the power of combining extractive and abstractive techniques through role- and context-aware processing. By classifying utterances based on dialogue acts (e.g., question, suggestion, decision), their model improved the quality of summaries, particularly in capturing meeting intent and actionable content. This approach aligns closely with the goals of action-item detection in long meeting transcripts, as understanding the speaker's intent and the surrounding context is crucial in distinguishing routine conversational exchanges from those that require follow-up or delegation.

3.4 Meeting Transcript & Corpora

The use of structured meeting transcripts plays an important role in understanding communication flow, speaker intent, and decision-making cues in professional environments. In the work by Zhong et al. (2021), the authors proposed a context-aware summarization model that segments meetings using a sliding window approach to maintain coherence and topic continuity. This approach enhances extractive summarization by capturing dynamic shifts in conversation, which is essential in real-life meeting scenarios where speakers often digress or loop back to earlier points. This study underlines the need for adaptive segmentation methods to improve downstream tasks like action-item detection and dialogue understanding—key elements in our proposed study.

Further, supervised machine learning approaches have been proven effective for textual tasks, especially when paired with annotated corpora. Kowsari et al. (2019) presented a comprehensive survey on supervised machine learning techniques for automatic text classification, emphasizing the role of high-quality training datasets and linguistic feature modeling in improving classification outcomes. Their review of classifiers, including support vector machines (SVMs), highlighted how context-sensitive learning combined with structured corpora can lead to robust text classifiers. This reinforces the decision to use Linear SVM in our system, where learning from annotated corpora like AMI and i-TANONG is vital for detecting and classifying actionable content in transcripts.

Meanwhile, recognizing the diversity in conversational behavior and local context has become increasingly relevant in corpus development. Blodgett et al. (2022) argued for using conversational corpora that capture nuances in speech, such as turn-taking, politeness strategies, and cultural expressions. They highlight how traditional NLP datasets often overlook the diversity of language use across communities. By leveraging culturally aware corpora like i-TANONG, which reflect Filipino discourse styles, our system aims to model and detect action items more accurately in meetings grounded in Philippine contexts—ensuring relevance, usability, and inclusivity.

Lastly, the MeetScript project by Ma et al. (2023) introduced innovative transcript-based interfaces that facilitate better engagement in group video meetings. Their system capitalized on real-time transcript interactions, allowing users to highlight, annotate, and navigate content seamlessly during meetings. This approach demonstrates the increasing relevance of intelligent transcript interfaces for hybrid work environments. It also supports the motivation behind our study: to not only detect action items but to potentially feed them into usable systems that enhance meeting productivity through intelligent summaries and task tracking based on localized and labeled corpora.

3.5 Sequence Labeling and Dialogue Act Recognition

Dialogue act recognition plays a critical role in understanding the structure and intent behind spoken utterances in meetings. It facilitates the identification of who spoke, when, and with what purpose—elements that are fundamental to action-item detection and meeting summarization. Recent advancements in natural language processing have focused on enhancing sequence labeling techniques to improve the recognition of speaker intent and conversational structure.

Kim et al. (2024) introduced Dual Process Masking, a cutting-edge approach to dialogue act recognition that independently learns from both utterance-level and contextual features, resulting in higher precision in complex conversational settings. Similarly, Chapuis et al. (2020) proposed a hierarchical pre-training strategy that captures dependencies at both the sentence and dialogue levels, improving classification performance in spoken dialog environments.

Raheja and Tetreault (2019) further advanced the field by developing a context-aware self-attention mechanism, which dynamically allocates attention based on surrounding dialogue turns, enabling more accurate recognition of speaker intent.

Meanwhile, Żelasko et al. (2021) highlighted the critical role of context, punctuation, and dialogue labels in helping transformer models effectively model conversational flow.

These contributions emphasize the value of incorporating conversational features such as speaker turns, discourse signals, and temporal sequencing—elements that are also leveraged in this study's SVM-based action-item extraction framework.

3.6 SVM-Based Techniques and Classical ML Models

Although deep learning dominates much of modern NLP, classical machine learning techniques—particularly Support Vector Machines (SVM)—remain highly relevant, especially in applications requiring interpretability, efficient training, and feature engineering. Classical models offer computational advantages and often perform competitively in structured domains like meeting analysis.

Abualigah et al. (2019) provided a foundational review of text summarization techniques, noting that while deep learning enables abstraction, classical methods excel in extraction-based tasks when paired with linguistic and positional features. Kumar et al. (2020) applied this hybrid approach in healthcare, demonstrating that ensembling classical and deep models can improve outcomes in tasks such as morbidity detection from clinical notes.

Kiminos and Gkaris (2020) explored the application of Naïve Bayes and SVM models in identifying decision points within business meeting transcripts. Their findings confirmed SVM's ability to detect structured, task-oriented segments within dialogue—a key requirement in action-item detection. Furthermore, Kadhim (2019) presented a comprehensive review of supervised learning approaches for text classification, reinforcing the robustness and adaptability of SVM in various NLP contexts.

Building upon these works, this study employs an SVM model enriched with linguistically-informed features, such as modal and imperative verbs, part-of-speech (POS) patterns, dependency relations, and sentence positioning, to classify action items within segmented meeting data. The result is a lightweight yet interpretable classifier suitable for real-time or resource-constrained environments.

3.7 Post-processing & Action-Item Extraction Techniques

Post-processing is essential for transforming raw predictions into coherent, human-readable outputs. In meeting summarization, this step ensures that extracted action items are properly contextualized and seamlessly integrated into the final summary. This contributes to usability by clarifying vague or incomplete utterances.

Pandya and Gawande (2022) developed an end-to-end system for the automatic generation of meeting minutes, emphasizing structured layout and clear delineation of decisions and action items. Vapnik and Izmailov (2021) introduced a Reinforced SVM model with memorization mechanisms, enabling models to retain and leverage prior conversational context—an essential capability in dialogues where action items are often distributed across multiple turns.

Sadia et al. (2025) extended these techniques into multilingual domains by applying large language models to action-item detection in Urdu meeting transcripts. Their work validates the versatility of hybrid models in diverse linguistic contexts. Similarly, Sachdeva et al. (2021) investigated pre-trained transformers for corporate meeting analysis, showing significant improvements in performance due to the incorporation of contextual embeddings.

Informed by these developments, this study introduces a contextual rephrasing mechanism that uses a fine-tuned BART model to re-summarize action items along with their surrounding sentences—defined as a "neighborhood." This step improves clarity and resolves pronouns or incomplete references, ensuring that each action item is self-contained, actionable, and aligned with the surrounding dialogue context.

3.8 Multimodal & Cultural Adaptation in Summarization

Multimodal aspects have confirmed improvements for automatic meeting summarization, since a multimodal automatic summarization system considers a variety of content, especially by looking at audio and video to determine the content's significance. For example, Nihei and Nakano (2019) designed a multimodal utterance-predictive model by identifying both verbal and nonverbal features (e.g., voice intensity, location of a speaker's gaze) to improve salient-sentence detection (F1 of 0.827). Their best performing deep-learning model outperformed models created by hand coding, showing meeting summarization model quality can be improved through multimodal studies.

From the results on cultural aspects of automatic summarization, training models for local languages and style is important for accessibility. An example of a language model with a cultural connotation is the iTANONG-DS dataset (Visperas et al., 2023), which includes labeled corpora of the Philippine languages Tagalog and Cebuano in NLP tasks like summarization. The iTANONG-DS dataset allows a developer or test researcher to train and evaluate models that understand syntactic structures of Philippine languages. As you mentioned, the use of code-mixing as an authentic local expression can assist with localized summarization performance for users in the videos.

Together, these findings provide research justification to pursue the multimodal design and localization to the Philippine context. By integrating audio/text features for action-item detection and language embeddings (fine-tuning or no hashing), the framework will be able to provide adequate summaries that are not just accurate, but in context with localization in mind for users in the meetings/contexts.

4 EQUATIONS AND MATHEMATICAL MODEL

The effectiveness of the proposed action-item detection system relies on the mathematical robustness of the Linear Support Vector Machine (SVM) and the efficiency of the feature hashing mechanism.

4.1 Linear SVM Formulation

The core objective of the Linear SVM is to find the maximum margin hyperplane that separates action items from information items in a high-dimensional feature space. Since meeting transcripts often contain overlapping linguistic patterns, we utilize a "soft margin" approach. The optimization problem is formulated as follows:

$$\min_{w,b} \quad \frac{1}{2} ||w||^2 + C \sum_{i=1}^{n} \quad \zeta_i$$

Subject to the constraint:

$$y_i(w \cdot x_i + b) \geq 1 - \zeta_i, \quad \zeta_i \geq 0$$

Where:

- $\boldsymbol{w}$ is the weight vector perpendicular to the hyperplane.
- $\boldsymbol{C}$ is the regularization parameter that balances the margin size against classification errors.
- $\boldsymbol{\zeta_i}$ (slack variables) allow for misclassifications in non-linearly separable data, which is common in diverse Taglish datasets.

4.2 Feature Hashing

To handle the massive scale of the 180,000-sample corpus without exhausting system memory, we employ Feature Hashing (the hashing trick). This mechanism maps arbitrary features (words or n-grams) into a fixed-size vector of 2^{10} (1,024) dimensions. The mapping function ϕ is defined as:

$$\phi(x) = \sum_{j:h(j)=i} \xi(j) x_j$$

Where:

- $\boldsymbol{h}$ is a hash function that maps a feature to an index in the range $[0, 2^{10} - 1]$
- ξ is a second hash function used to determine the sign of the feature value, helping to mitigate the impact of hash collisions.
- This ensures that the model remains lightweight and capable of incremental learning via SGDClassifier.

4.3 Decision Function

Once the optimal weight vector w and bias b are determined during the training phase, the model classifies a new sentence x from a transcript using the following decision function:

$$f(x) = \text{sign}(w \cdot x + b)$$

The output is binary:

- **+1 (Action Item):** If the calculated value is positive, the sentence is flagged as a task.
- **-1 (Information Item):** If the value is negative, the sentence is treated as a general statement or informational context.

5 METHODOLOGY & FRAMEWORK

This chapter details the systematic six-phase methodology used to develop an interpretable system for action-item detection and meeting summarization. The design focuses on transforming meeting transcripts into formal Minutes of the Meeting (MoM) report by combining Linear SVM for precise task extraction with BART-based abstractive summarization. By bridging the gap between human dialogue and professional documentation, this pipeline ensures that the final output is both computationally accurate and operationally useful for academic and professional institutions.

5.1 Data Acquisition & Preprocessing

This phase focuses on assembling and preparing a high-quality, culturally relevant dataset to train the action-item detection model. Two primary sources are utilized: the AMI Meeting Corpus, which contains English transcripts from real-world team meetings with speaker turns, roles, and dialogue annotations; and the iTANONG dataset, a Taglish (Tagalog-English) open-domain dialogue dataset. The AMI dataset provides structured meeting-style discourse essential for training general action-item classifiers, while the iTANONG dataset brings in cultural and linguistic diversity reflective of Filipino communication patterns. Sentences from both datasets are manually annotated as either “Action Item” or “Not an Action Item” using custom guidelines. For the iTANONG dataset in particular, the focus is on identifying indirect task delegation,

politeness markers (e.g., “po,” “baka pwede”), and volunteer phrases (e.g., “sige, ako na lang”). Each annotated sentence is enriched with metadata such as speaker role, sentence position within the turn, use of honorifics, and Taglish detection. The resulting structured dataset combines cross-cultural examples and supports the training of a Linear SVM model that generalizes across both formal and culturally grounded communication, making the system adaptable for Philippine academic and organizational use.

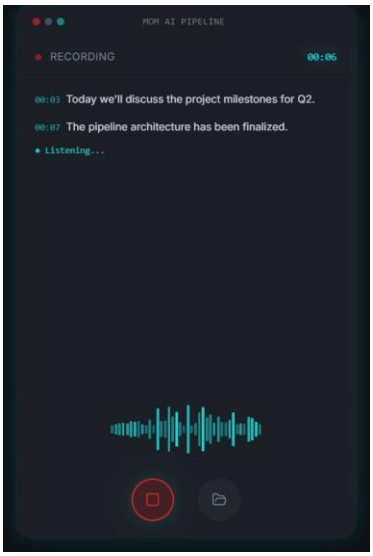

Figure 1: System Recording/Transcribing

5.2 Topic Segmentation

Long meeting transcripts are difficult to summarize holistically due to their length and varied content. This phase divides transcripts into smaller, coherent topic-based chunks using topic segmentation algorithms adapted from Golia and Kalita (2023). Three segmentation strategies are considered: Chunked Linear, Simple Cosine Similarity, and Complex Cosine Similarity, all leveraging MPNet embeddings to compute semantic similarity between speaker turns. By segmenting the transcript into manageable units, the system ensures that subsequent action item detection and summarization tasks remain context-aware and computationally feasible. This segmentation also preserves conversational flow and thematic continuity, enabling more accurate extraction of action-relevant content.

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PHASE 2: TOPIC SEGMENTATION

[SEGMENT 1]:
Alright, everyone Before you head out, I want to make sure we're all on the same page regarding your responsibilities for our next session I've just post
ed the materials for Chapter 7 online, please review the lecture slides and the supplementary video before Thursday's lab, as we'll be jumping straight i
nto the practical application Your main priority is Problem Set 3, which is now live on the portal

[SEGMENT 2]:
It consists of five calculation-based questions and I expect to see your full step-by-step logic, rather than just the final answers if you want to earn
full credit Additionally, make sure you've read the Sorensen article on fluid mechanics Pay close attention to the second half and come prepared with at
least one specific question for our opening discussion Your problem sets are due by midnight this Monday, and if you run into any major roadblocks, I'll
be in my office tomorrow at 2 p to help you work through them

[SEGMENT 3]:
That's all for today Have a productive afternoon!
```

Figure 2: Topic Segmentation

5.3 Extractive Action Item & Opinion Classification using Linear SVM

At the core of the thesis is the detection of action items using a Linear Support Vector Machine (SVM) classifier enhanced by linguistically engineered features. For each sentence within a topic chunk, a set of lexical, syntactic, speaker-based, and culturally grounded features are extracted. These features include the presence of action verbs, modal auxiliaries, task-deadline phrases, honorifics, and indicators of Filipino-style politeness and indirect delegation. The Linear SVM is trained on the manually annotated dataset from Phase 1 to distinguish between action items and non-action items. This interpretable machine learning approach allows for transparency in classification decisions and supports the cultural localization of the model. Identified action items are then contextualized using their neighboring sentences, preparing them for integration into the final summary.

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PHASE 3: ACTION ITEM EXTRACTION & SELF-TRAIN

  [ ] Info: Alright, everyone
  [ ] Info: Before you head out, I want to make sure we're all on the same page regarding your responsibilities for our next session
  [!] Action: I've just posted the materials for Chapter 7 online,
  [!] Action: please review the lecture slides and the supplementary video before Thursday's lab, as we'll be jumping straight into the practical applica
tion
  [!] Action: Your main priority is Problem Set 3, which is now live on the portal
  [!] Action: It consists of five calculation-based questions and I expect to see your full step-by-step logic, rather than just the final answers if you
 want to earn full credit
  [!] Action: Additionally, make sure you've read the Sorensen article on fluid mechanics
  [!] Action: Pay close attention to the second half and come prepared with at least one specific question for our opening discussion
  [!] Action: Your problem sets are due by midnight this Monday, and if you run into any major roadblocks, I'll be in my office tomorrow at 2 p
  [ ] Info: to help you work through them
  [ ] Info: That's all for today
  [ ] Info: Have a productive afternoon!
```

Figure 3: Action Item Extraction

5.4 Abstractive Summarization

This phase focuses on generating readable and informative summaries that not only condense the transcript content but also integrate detected action items. A fine-tuned BART (Bidirectional and Auto-Regressive Transformer) model, pre-trained on conversational summarization datasets such as SAMSUM and XSUM, is used to produce general summaries for each topic-based chunk. Once action items are identified and contextualized, they are appended to the chunk summary, and the combined text is passed through the BART model again to ensure coherence and fluency. This dual-stage summarization ensures that actionable content is not only preserved but also highlighted within a natural-language summary, resulting in summaries that are more meaningful, task-oriented, and usable.

```
PHASE 4: SUMMARIZATION (BART)

--- Summary for Segment 1 ---
INPUT: Alright, everyone Before you head out, I want to make sure we're all on the same page regarding your...
OUTPUT: Review lectures and supplementary video for chapter 7 before lab on Thursday. Main priority is problem set 3, now live online on portal.

--- Summary for Segment 2 ---
INPUT: It consists of five calculation-based questions and I expect to see your full step-by-step logic, ra...
OUTPUT: Your problem sets will be due by Midnight this Monday. I expect you to see full step by step logic, not just final answers. Make sure you have re
ad the article about fluid mechanics in the first half of the task.

--- Summary for Segment 3 ---
INPUT: That's all for today Have a productive afternoon!...
OUTPUT: Me meeting: That is all for now. Have aproductive afternoon.
```

Figure 4: Abstractive Summarization

5.5 Recursive Merging & Minutes of the Minutes (MoM) Construction

To create a unified record for the entire meeting, this phase merges all topic-level summaries into a structured, comprehensive document. If the combined output remains within the token limits of the summarization model (e.g., BART's 1024-token input limit), it is processed through a final refinement layer to improve cohesion. Once the content is within this limit, the system automatically generates a formal Minutes of the Meeting (MoM) output. This structured final document organizes the refined text into professional sections, including meeting metadata, discussion highlights, and a consolidated list of the action items detected in Phase 3. If the initial combined text is too large, a recursive summarization strategy is employed—treating the concatenated summaries as a new document and reapplying the process until the content is condensed enough to be formatted into the final MoM. This ensures the final output is not just a summary, but a professional-grade record that is concise, coherent, and operationally useful.

5.6 Evaluation & Metrics

To validate system performance and demonstrate its real-world usability, this phase includes both quantitative evaluation and model explainability. The action-item classifier is assessed using metrics such as accuracy, precision, recall, and F1-score, while the quality of the summaries is measured using ROUGE and BERTScore. In addition, the interpretability of the Linear SVM model is highlighted by analyzing feature importance weights—demonstrating which linguistic or cultural features most influence action-item detection. This level of transparency fosters trust in the system and ensures its outputs are verifiable and aligned with human expectations, especially in culturally nuanced contexts like Filipino meetings.

5.7 THEORETICAL FRAMEWORK

The theoretical framework of this study centers on the belief that effectively identifying action items in lengthy transcripts requires a thoughtful balance between advanced AI and traditional, transparent machine learning. By rooting the system in Linguistic Feature Modeling, the research suggests that tasks aren't just random words but are signaled by specific human patterns, such as the use of "need to," "should," or direct command verbs. To ensure this process isn't just a "black box," the framework utilizes Explainable AI through a Linear SVM, allowing us to see exactly which words or markers triggered a decision. This is especially vital for Cultural Adaptation, as it accounts for the unique way Filipinos delegate tasks through Taglish and politeness markers like "paki" or "po." Finally, the framework employs Recursive Processing and Contextual Rephrasing to transform these identified items into a structured Minutes of the Meeting (MoM), ensuring that "buried" tasks are extracted along with their responsible parties and deadlines into a clear, actionable format.

5.8 CONCEPTUAL FRAMEWORK

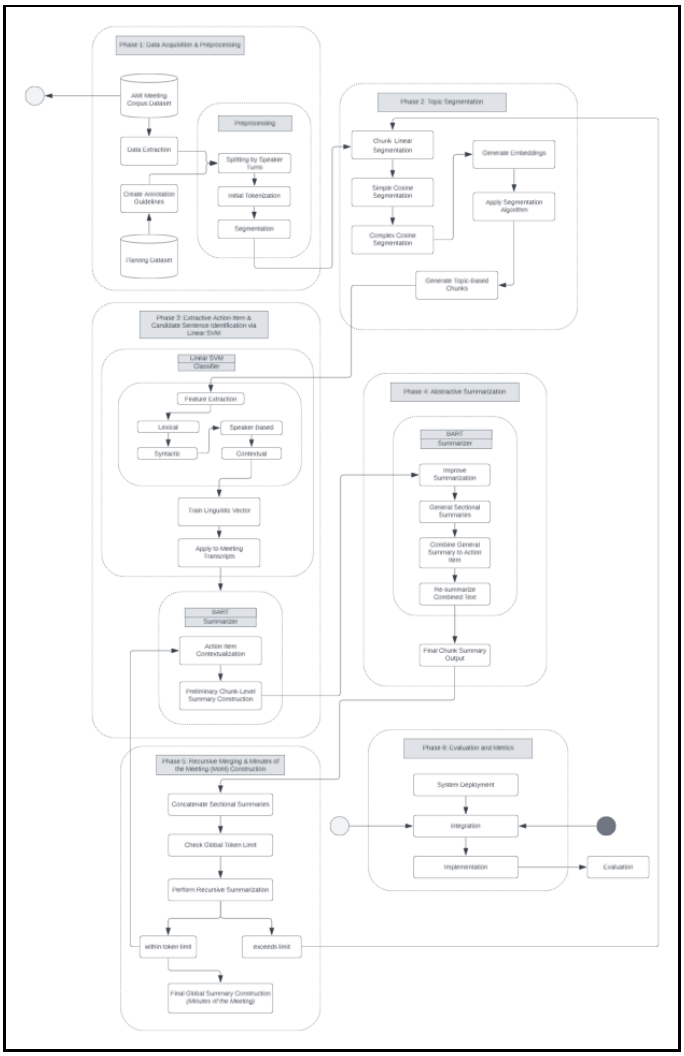

Figure 5: Conceptual Framework of the Study

6 RESULTS & DISCUSSION

This section evaluates the performance of the Linear SVM across multiple domains and data scales, emphasizing the model's predictive accuracy and its ability to provide interpretable insights into meeting dynamics.

6.1 Performance Metrics

The model was evaluated using a held-out test set from each specialized domain. The primary metrics used were Precision (the accuracy of task flagging), Recall (the ability to find all tasks), and the F1-Score (the harmonic mean of the two).

Table 1: Evaluation Metrics Across Specialized Linguistic Domains

Dataset Context	Accuracy	Precision	Recall	F1-Score
Meeting Specific	89.34%	0.84	0.98	0.90
Formal English	83.18%	0.75	0.99	0.86
Comprehensive Thesis	86.44%	0.87	0.86	0.86
Expanded Context	86.27%	0.79	0.98	0.88
Mixed Taglish	73.43%	0.70	0.82	0.75

The results indicate that the model excels in Meeting Specific contexts, achieving its highest F1-score of 0.90. Notably, the model demonstrates very high Recall (up to 0.98) across most sets, ensuring that very few actionable tasks are missed. The lower accuracy in the Mixed Taglish set (73.43%) reflects the inherent linguistic noise and non-standard syntax common in informal code-switching.

6.2 Interpretability Analysis

By utilizing a Linear SVM, we can interpret the classification logic by analyzing the coefficients of the weight vector w. This allows us to identify the specific linguistic markers that the model associates with actionable intent.

Table 2: Top 10 High-Weight Features for Action Item Classification

Rank	Feature (Word)	Weight	Context/Linguistic Role
1	paki	+5.42	Taglish Politeness/Imperative Prefix
2	submit	+4.81	English Action Verb
3	need to	+4.33	Modal of Obligation
4	deadline	+4.15	Temporal Constraint
5	finalize	+3.98	Project Conclusion Marker
6	ihanda	+3.72	Tagalog Imperative (Prepare)
7	assignment	+3.55	Academic Task Marker
8	Schedule	+3.21	Planning/Organizational Verb
9	i-check	+3.10	Code-switched Action Verb
10	ASAP	+2.98	Urgency Indicator

The analysis confirms that the model heavily relies on both standard English imperatives (*"submit"*) and Filipino-specific markers (*"paki-"*). The high weight of "paki-" validates the necessity of including culturally grounded features for models deployed in the Philippine professional sector.

6.3 Comparative Analysis: Data Scaling and Linguistic Variety

Comparing the performance across the different datasets reveals how data scale and language type affect the SVM's decision boundary:

1. **Impact of Scale:** Increasing the data from the 10k Formal English set to the 20k Expanded Contexts set improved the model's ability to distinguish "Information" from "Action Items," raising the F1-score from 0.86 to 0.88 despite the increased complexity of the text.
2. **Linguistic Challenges:** There is a significant performance gap (approx. 16%) between Meeting Specific (89.43%) and Mixed Taglish (73.43%). This suggests that while the SVM is highly effective for structured professional

dialogue, informal Taglish requires more intensive feature engineering to account for inconsistent grammar and slang.

3. **High Recall Strategy:** Across all datasets, the model consistently prioritized Recall over Precision for Action Items. For the Expanded Contexts, the recall reached 0.98, indicating the model is optimized to "over-capture" potential tasks rather than risk missing a commitment made during a meeting.

7 CONCLUSION AND RECOMMENDATIONS

This study successfully demonstrated the viability of an interpretable, scalable framework for action-item detection within multilingual (Taglish) meeting transcripts. By prioritizing linguistic feature modeling and a Linear SVM over high-resource deep learning, the research provides a computationally efficient and transparent solution for administrative automation in the Philippine context.

7.1 Summary of Findings

The experimental results validate the effectiveness of the Linear Support Vector Machine (SVM) when trained on diverse datasets totaling over 180,000 samples. Key findings include:

- **High Recall Performance:** Across all datasets, including Formal English (0.99) and Expanded Contexts (0.98), the model demonstrated exceptional recall. This ensures that the system captured nearly all actionable commitments, a critical requirement for reliable Minutes of the Meeting (MoM) generation.
- **Linguistic Resilience:** The model maintained a robust F1-score of 0.89 to 0.90 in meeting-specific and comprehensive contexts. While accuracy naturally dipped to 73.43% in heavily mixed Taglish sets, the model remained effective at identifying tasks despite non-standard syntax.
- **Scalability through Incremental Learning:** The implementation of SGDClassifier and Feature Hashing allowed for the processing of massive data scales without specialized GPU hardware. This proves that high-performing NLP tools can be deployed on standard office hardware.
- **Cultural Interpretability:** Feature importance analysis confirmed that Filipino-specific markers, such as the prefix "paki-" and code-switched verbs like "i-check," are the primary drivers of classification. This transparency allows users to verify the "logic" behind every flagged task, distinguishing this approach from "black-box" neural networks.

7.2 Future Work

Based on the performance metrics and observations during development, the following areas are recommended for future research:

- **Live Transcription Integration:** Future iterations should transition from asynchronous file processing to Real-time Speech-to-Text (STT) integration. This would enable the system to provide immediate notifications to participants as tasks are assigned during a live session.
- **Hybrid Semantic Modeling:** To bridge the accuracy gap in complex Taglish dialogue, researchers may explore a hybrid architecture. A Linear SVM can be used for high-speed keyword extraction, while a lightweight transformer like DistilBERT could resolve ambiguous or "indirect" tasks that lack explicit imperative markers.

- **Expansion of Regional Dialects:** Given the success of the Taglish model, the framework should be expanded to include other major Philippine languages, such as Cebuano and Ilocano, ensuring the tool's utility across the entire archipelago.
- **Multi-modal Sentiment Analysis:** Incorporating prosodic features (tone of voice) could help the model distinguish between a casual suggestion and a formal directive, further refining the precision of the extraction process.

ACKNOWLEDGMENTS

With heartfelt appreciation, we wish to acknowledge the individuals whose support was significant to the success of this research.

First and foremost, we offer our deepest thanks to the Almighty God for His divine guidance and for granting us the clarity and perseverance required at every stage of this study.

We are truly grateful to Dr. Marc P. Laureta, our thesis adviser, for his expert mentorship and constant encouragement. His dedication and insightful advice were the driving forces behind the completion of this work.

We also want to extend our heartfelt thanks to our classmates and fellow 4BSCS-1 members. Their teamwork, constant encouragement, and friendship helped us overcome the challenges we encountered along the way.

To our parents, we express our love and gratitude for your immense patience and for supporting us emotionally and financially. Your sacrifices provided the foundation for our success. Ultimately, we offer our highest praise to God, through whom all things are made possible.

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